



```
In [2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [3]: titanic = sns.load_dataset('titanic')
```

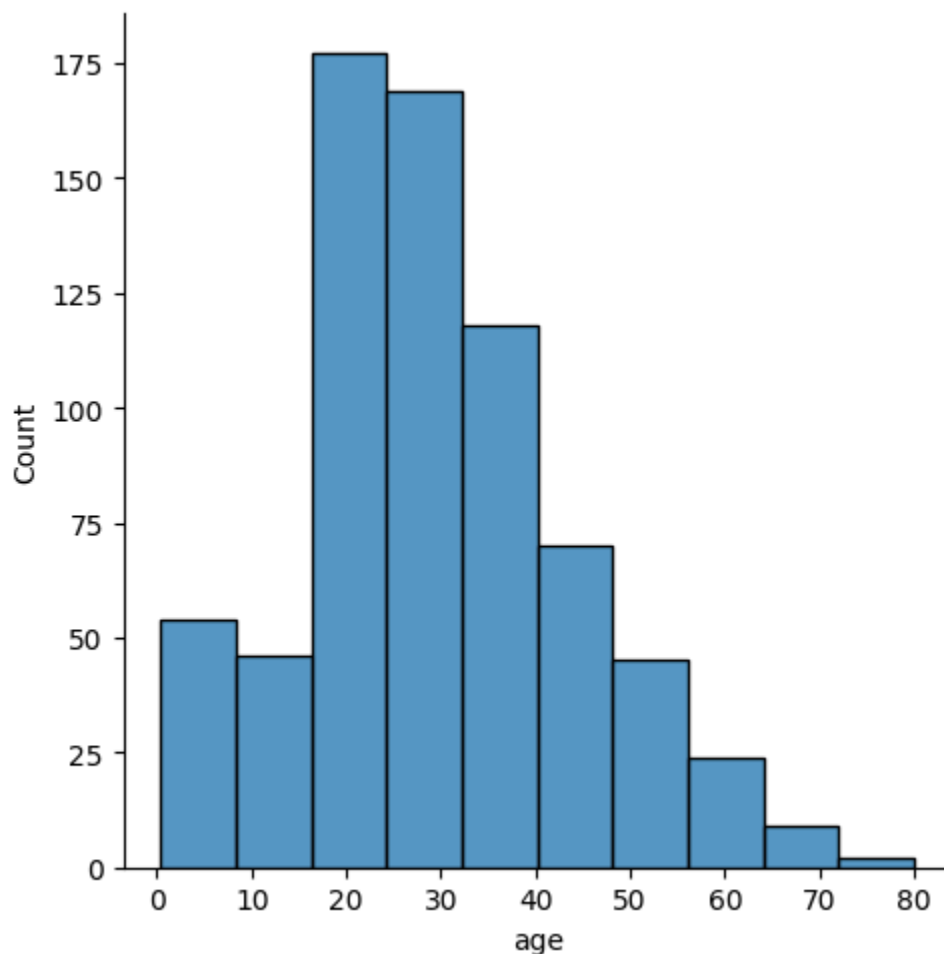
```
In [4]: titanic.head()
```

```
Out[4]:
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who
0	0	3	male	22.0	1	0	7.2500	S	Third	mar
1	1	1	female	38.0	1	0	71.2833	C	First	womar
2	1	3	female	26.0	0	0	7.9250	S	Third	womar
3	1	1	female	35.0	1	0	53.1000	S	First	womar
4	0	3	male	35.0	0	0	8.0500	S	Third	mar

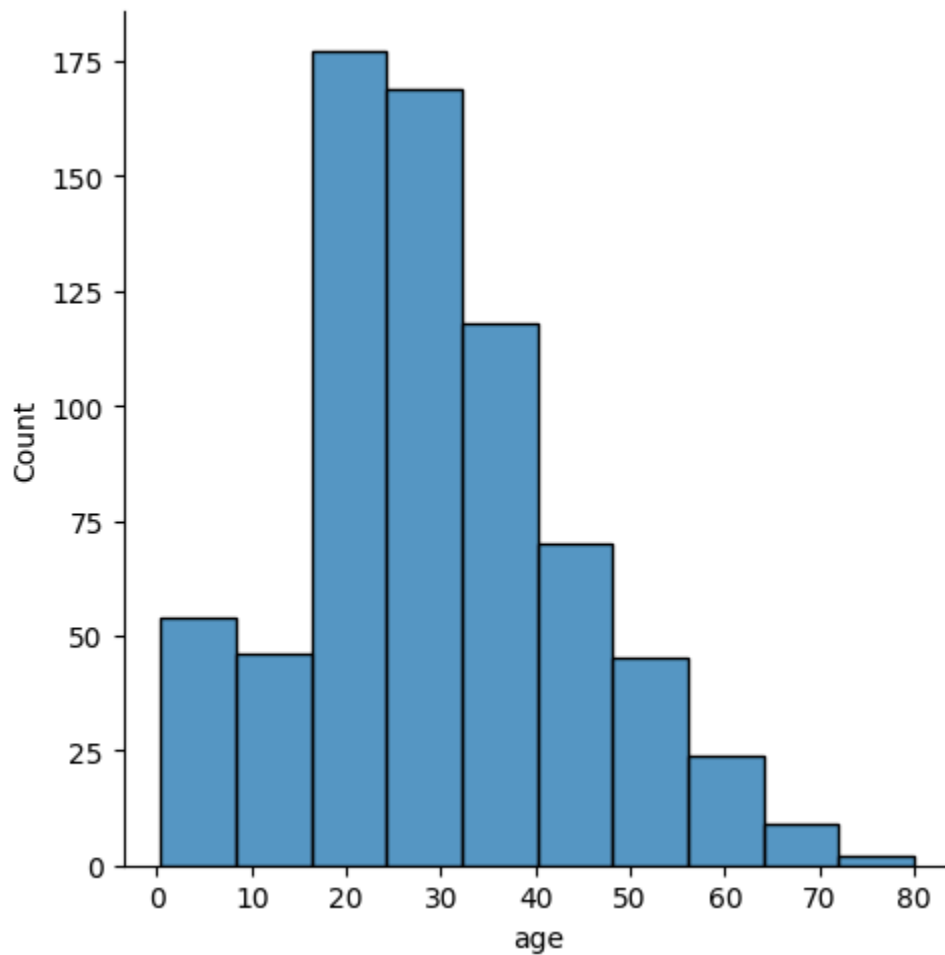
```
In [5]: sns.displot(titanic['age'], bins = 10)
```

```
Out[5]: <seaborn.axisgrid.FacetGrid at 0x2aeffd9a0f0>
```



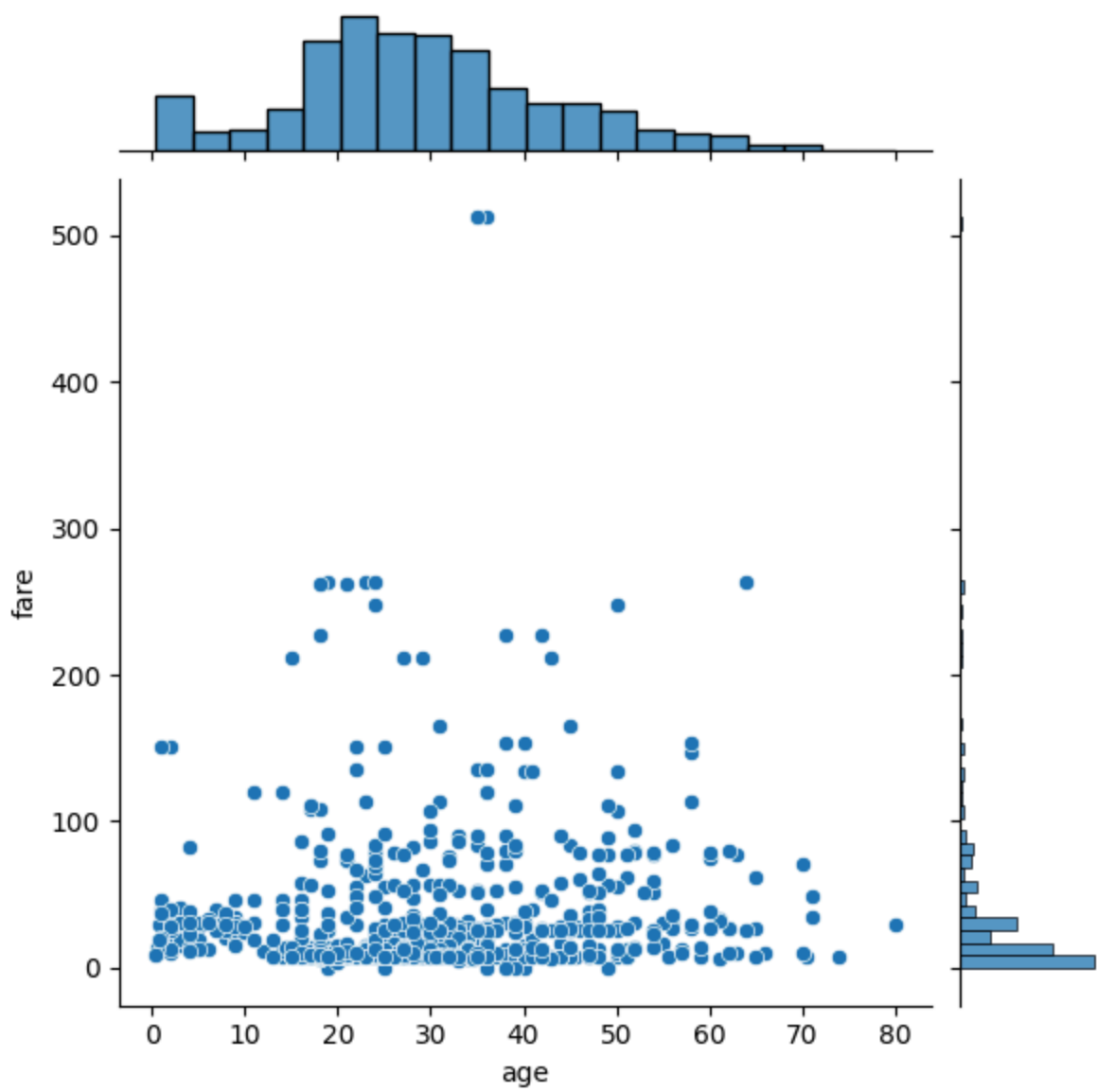
```
In [6]: sns.displot(titanic['age'], bins = 10, kde = False)
```

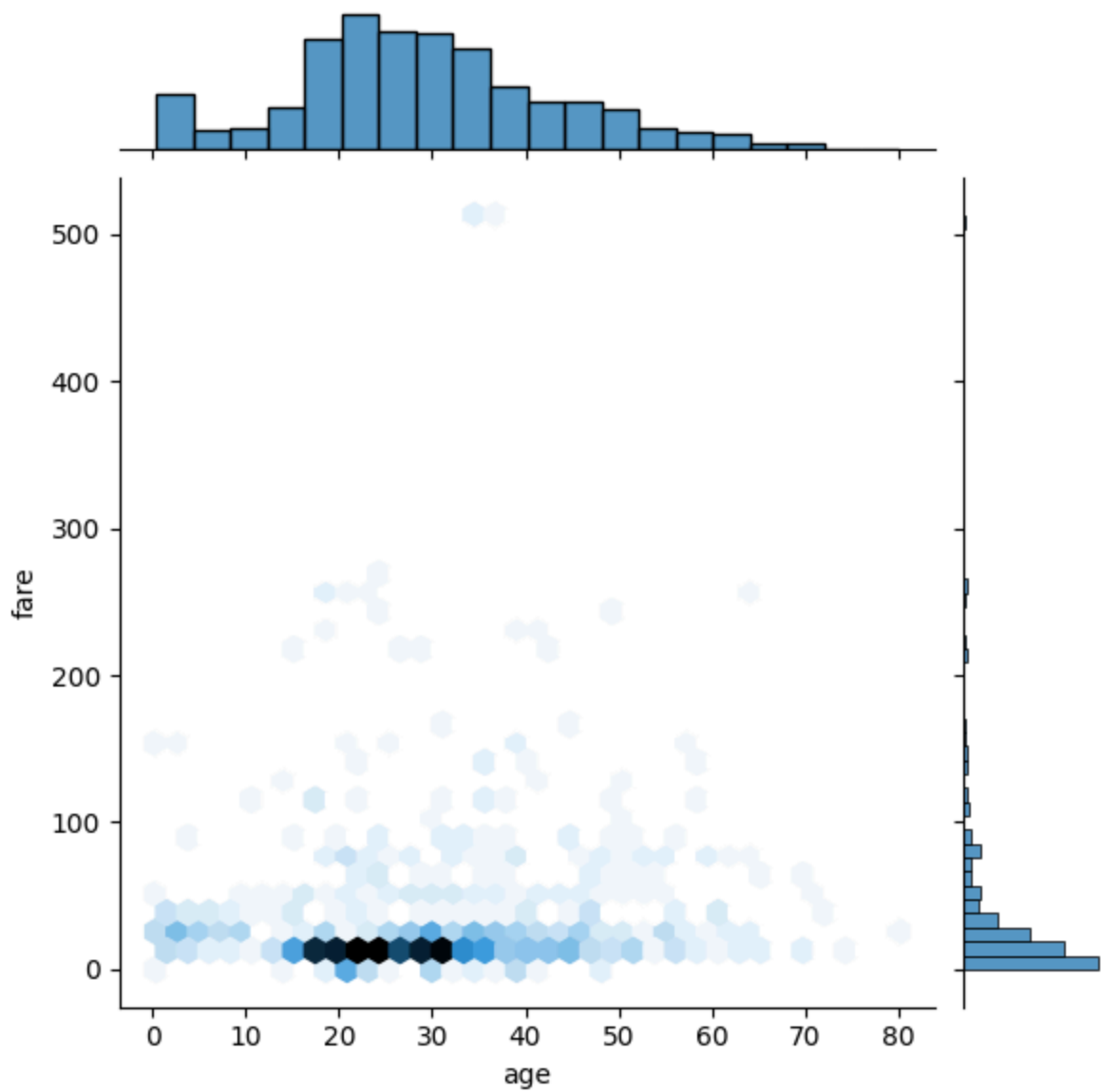
```
Out[6]: <seaborn.axisgrid.FacetGrid at 0x2ae8162e660>
```



```
In [7]: sns.jointplot(x = titanic['age'], y = titanic['fare'], kind = 'scatter')
sns.jointplot(x = titanic['age'], y = titanic['fare'], kind = 'hex')
```

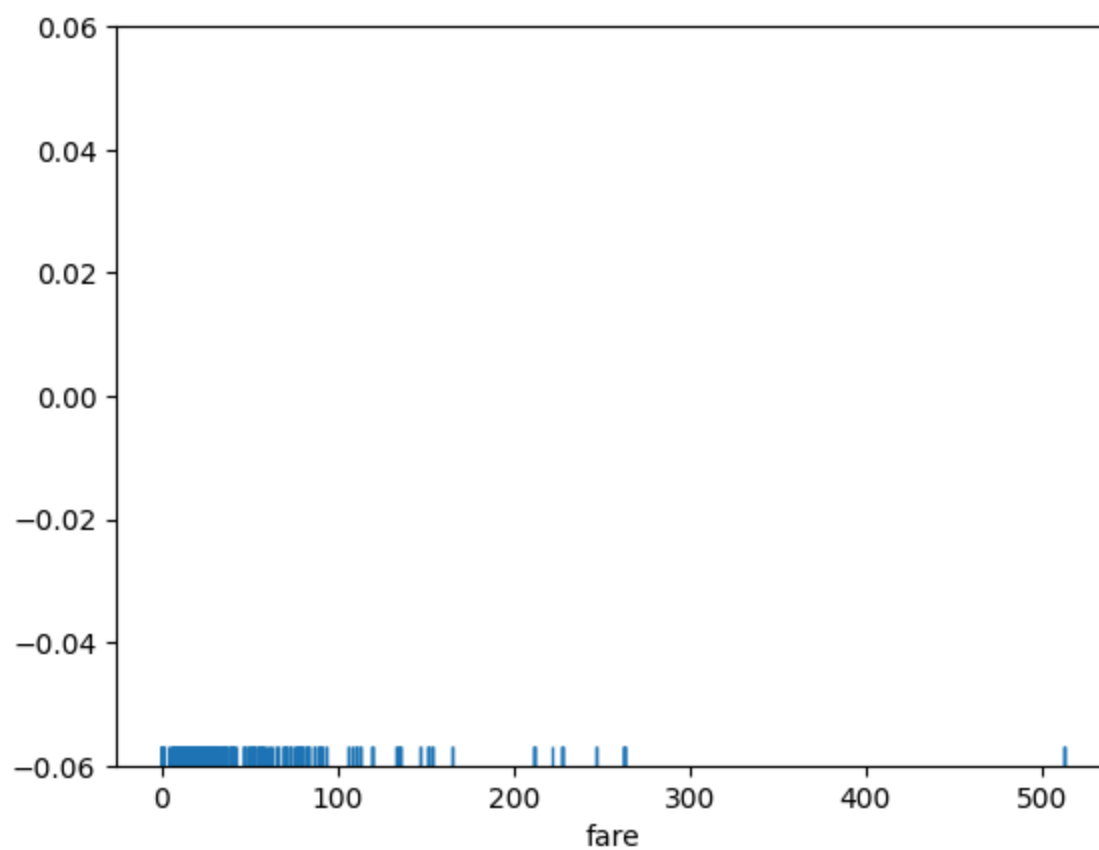
```
Out[7]: <seaborn.axisgrid.JointGrid at 0x2ae81801d00>
```





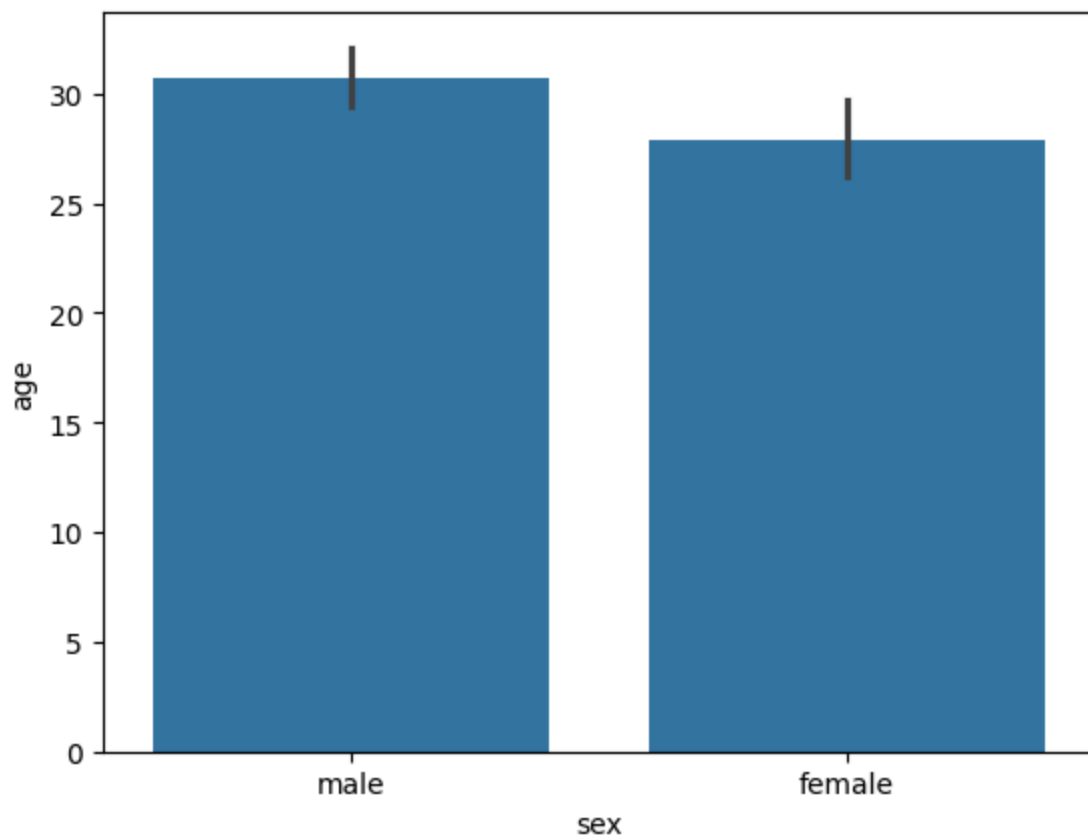
```
In [8]: sns.rugplot(titanic['fare'])
```

```
Out[8]: <Axes: xlabel='fare'>
```



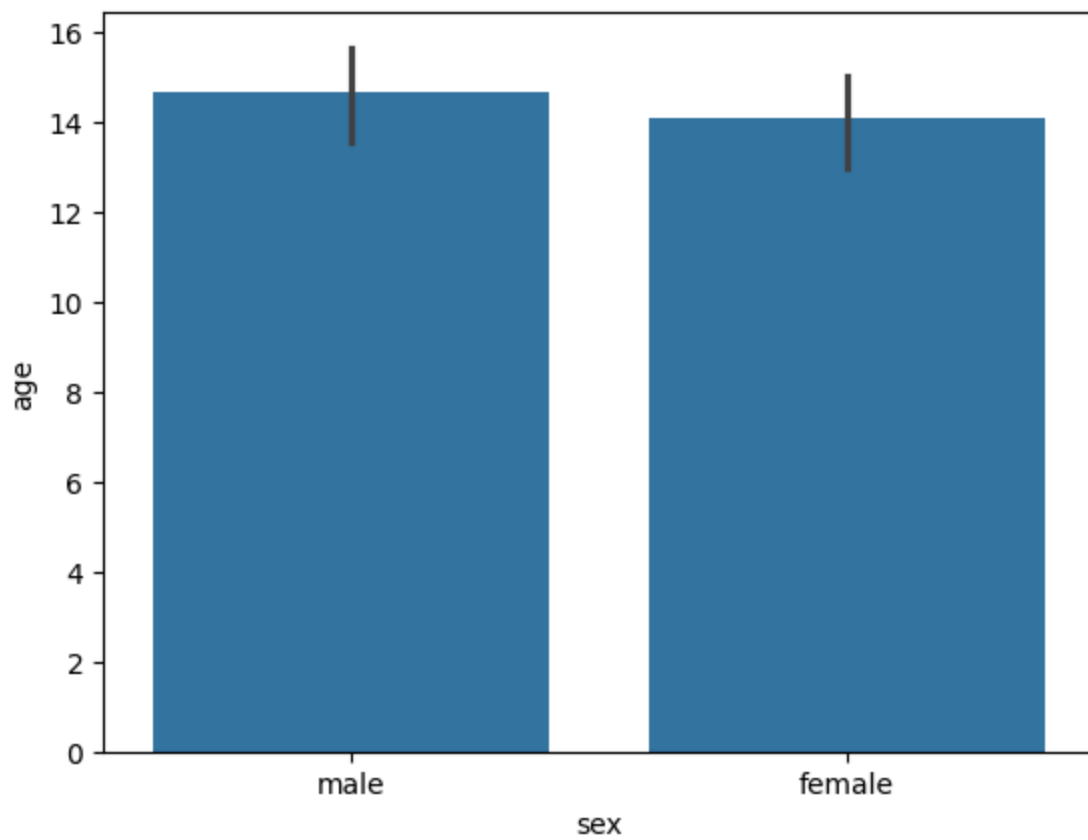
```
In [9]: sns.barplot(x = 'sex', y = 'age', data = titanic)
```

```
Out[9]: <Axes: xlabel='sex', ylabel='age'>
```



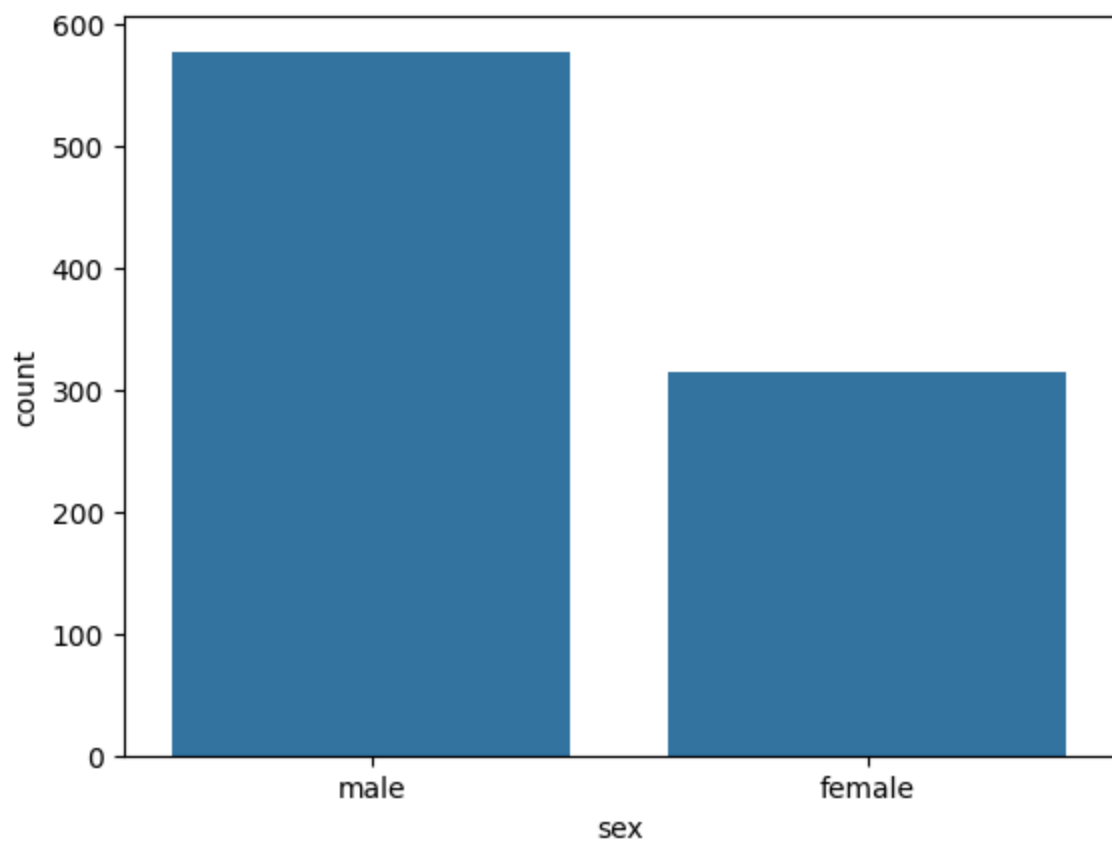
```
In [10]: sns.barplot(x = 'sex', y = 'age', data = titanic, estimator = np.std)
```

```
Out[10]: <Axes: xlabel='sex', ylabel='age'>
```



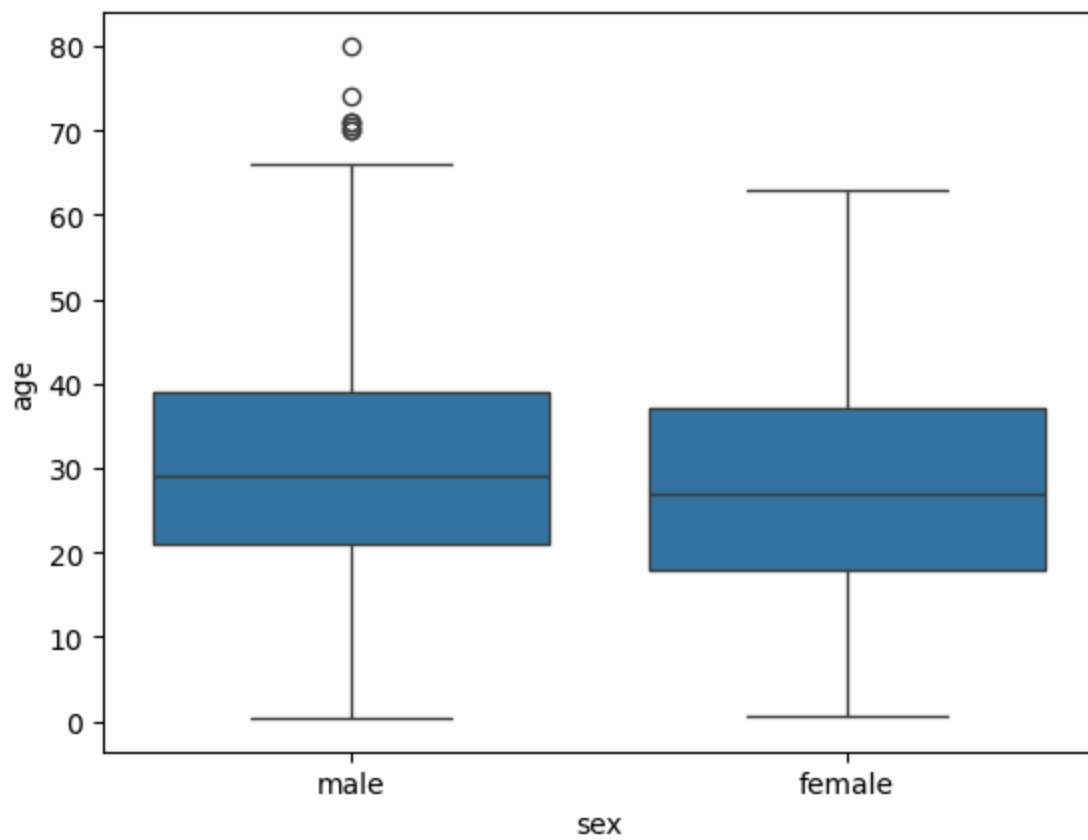
```
In [11]: sns.countplot(x = 'sex', data = titanic)
```

```
Out[11]: <Axes: xlabel='sex', ylabel='count'>
```



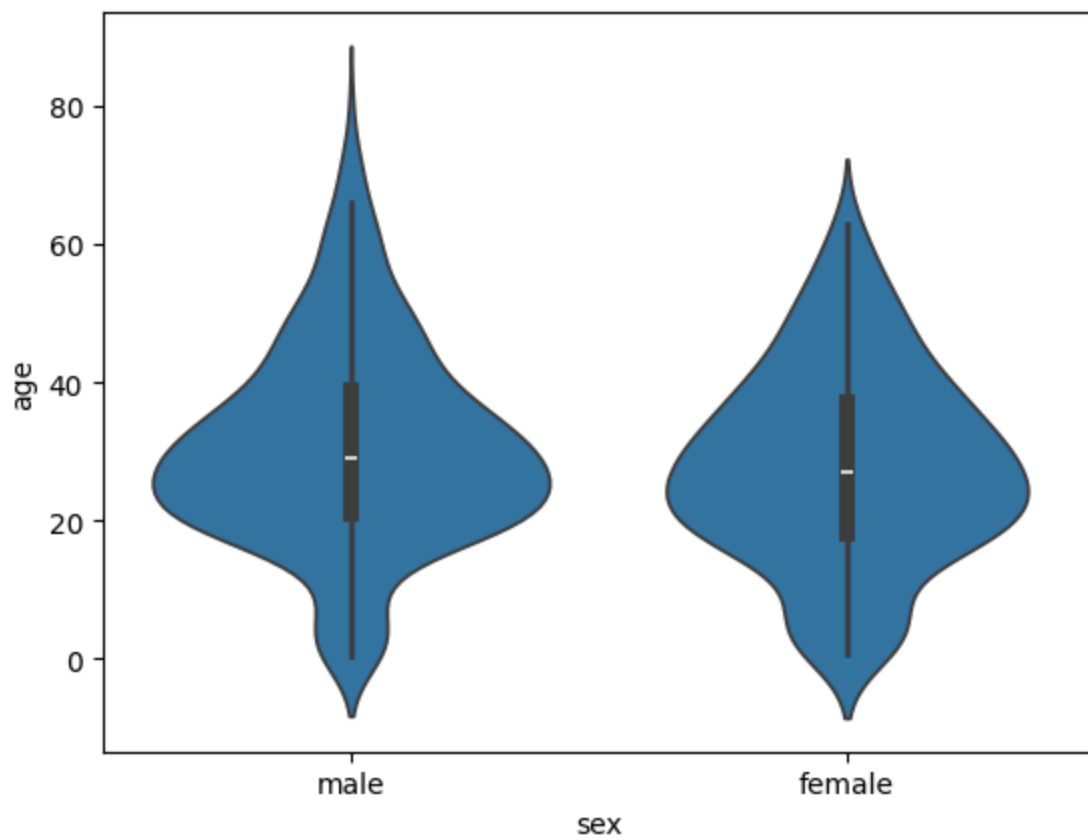
```
In [12]: sns.boxplot(x = 'sex', y = 'age', data = titanic)
```

```
Out[12]: <Axes: xlabel='sex', ylabel='age'>
```

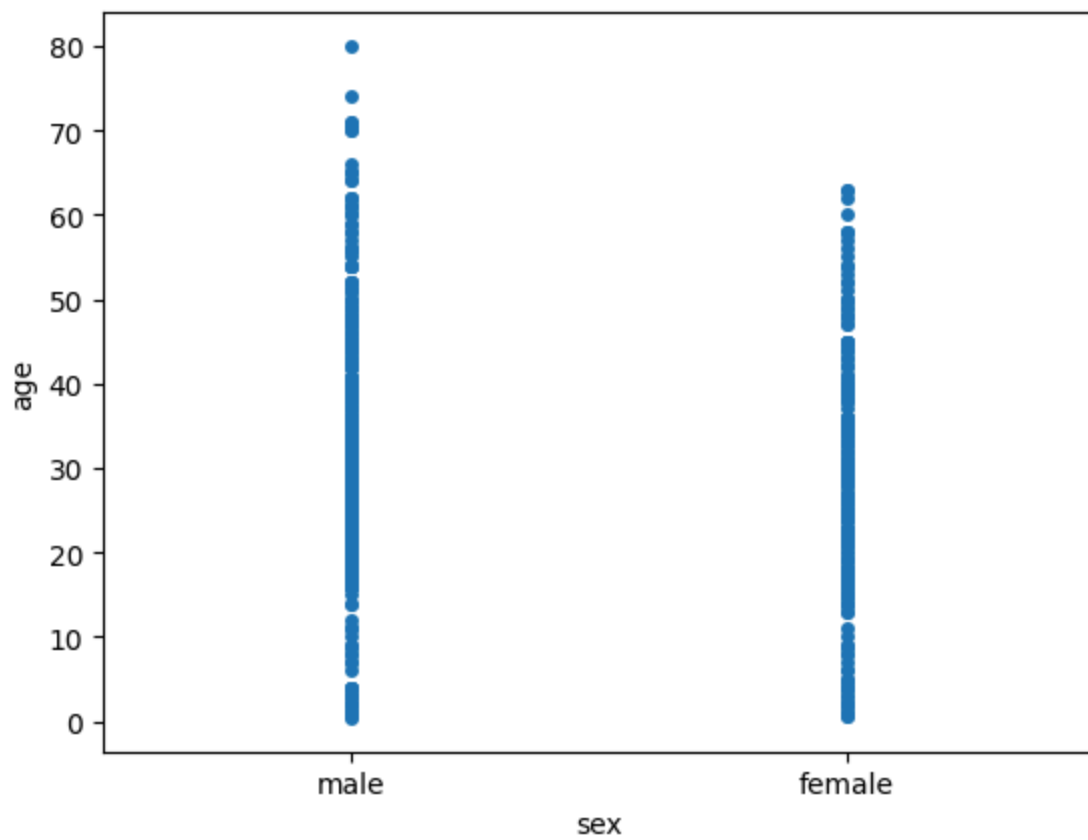
```
In [13]: sns.violinplot(x = 'sex', y = 'age', data = titanic)
```

```
Out[13]: <Axes: xlabel='sex', ylabel='age'>
```



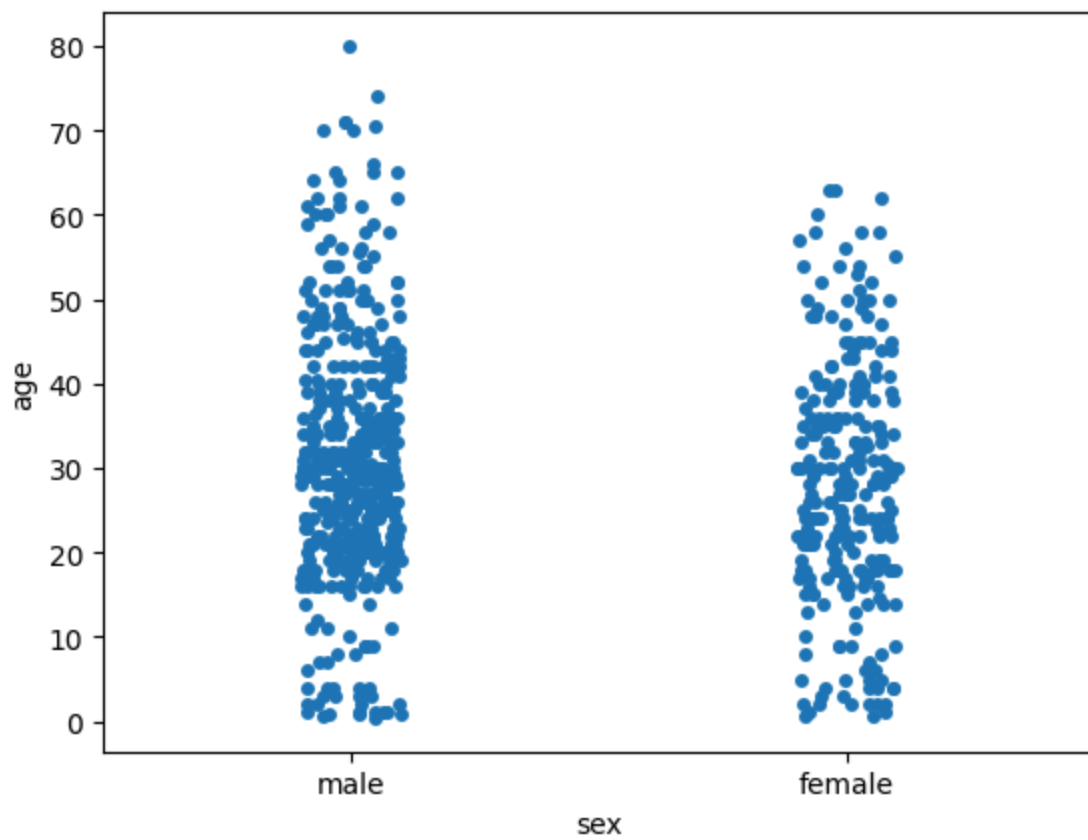
```
In [14]: sns.stripplot(x = 'sex', y = 'age', data = titanic, jitter=False)
```

```
Out[14]: <Axes: xlabel='sex', ylabel='age'>
```



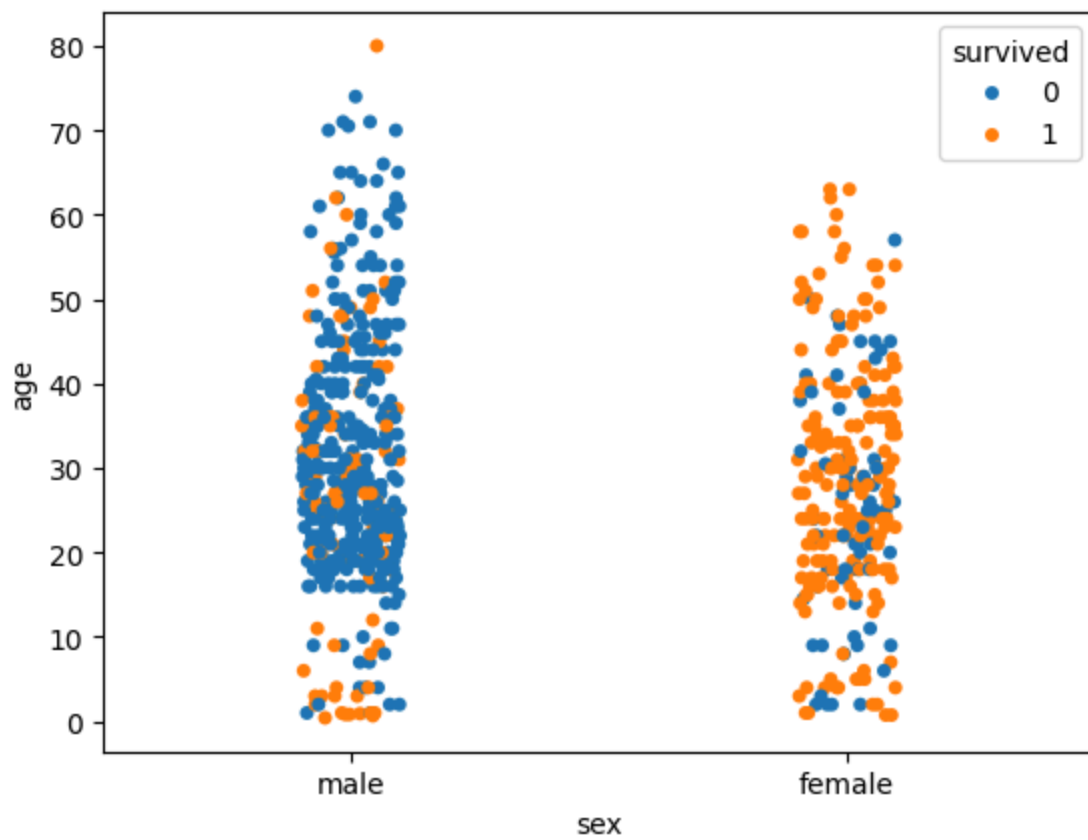
```
In [15]: sns.stripplot(x = 'sex', y = 'age', data = titanic, jitter=True)
```

```
Out[15]: <Axes: xlabel='sex', ylabel='age'>
```



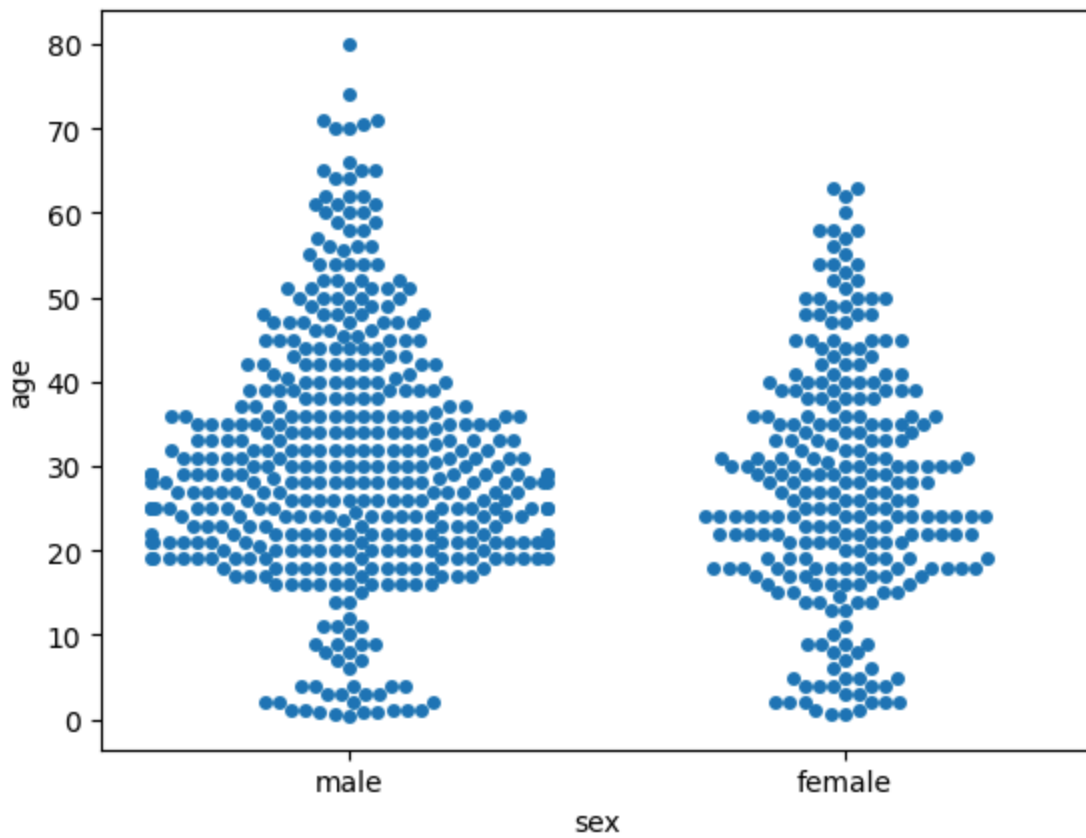
```
In [16]: sns.stripplot(x = 'sex', y = 'age', data = titanic, jitter=True, hue = 'surviv
```

```
Out[16]: <Axes: xlabel='sex', ylabel='age'>
```



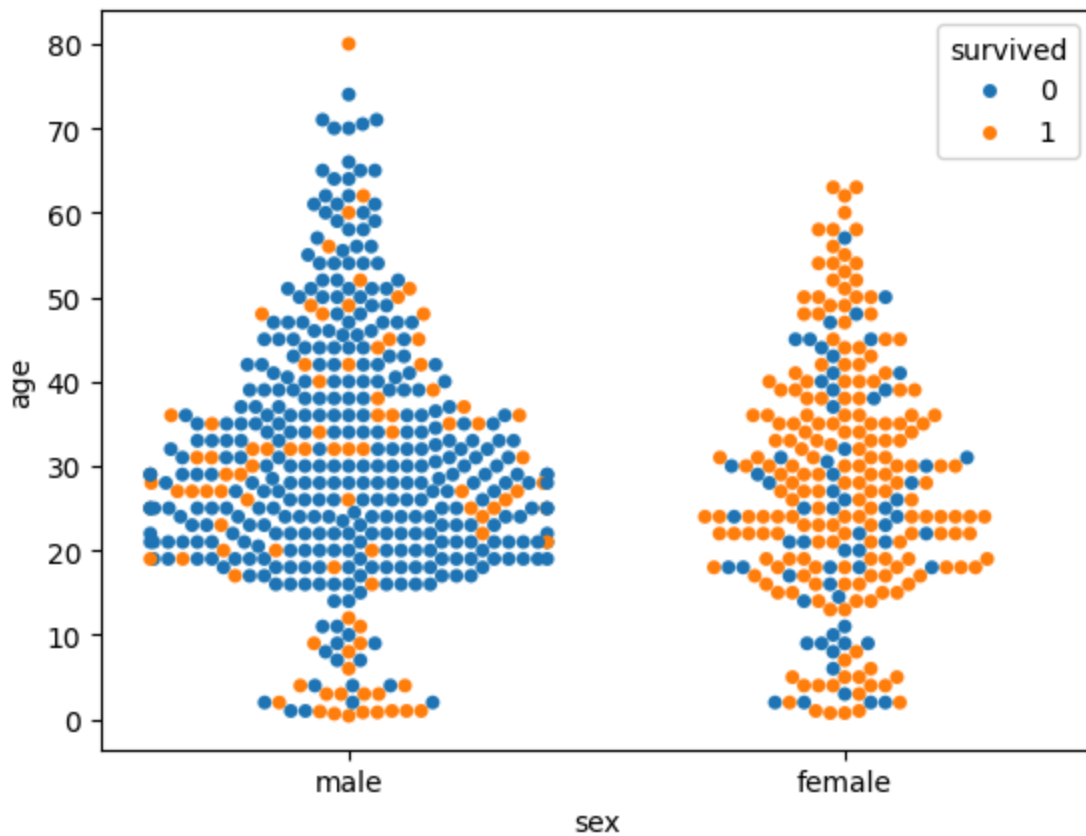
```
In [17]: sns.swarmplot(x = 'sex', y = 'age', data = titanic)
```

```
Out[17]: <Axes: xlabel='sex', ylabel='age'>
```



```
In [18]: sns.swarmplot(x = 'sex', y = 'age', data = titanic, hue = 'survived')
```

```
Out[18]: <Axes: xlabel='sex', ylabel='age'>
```



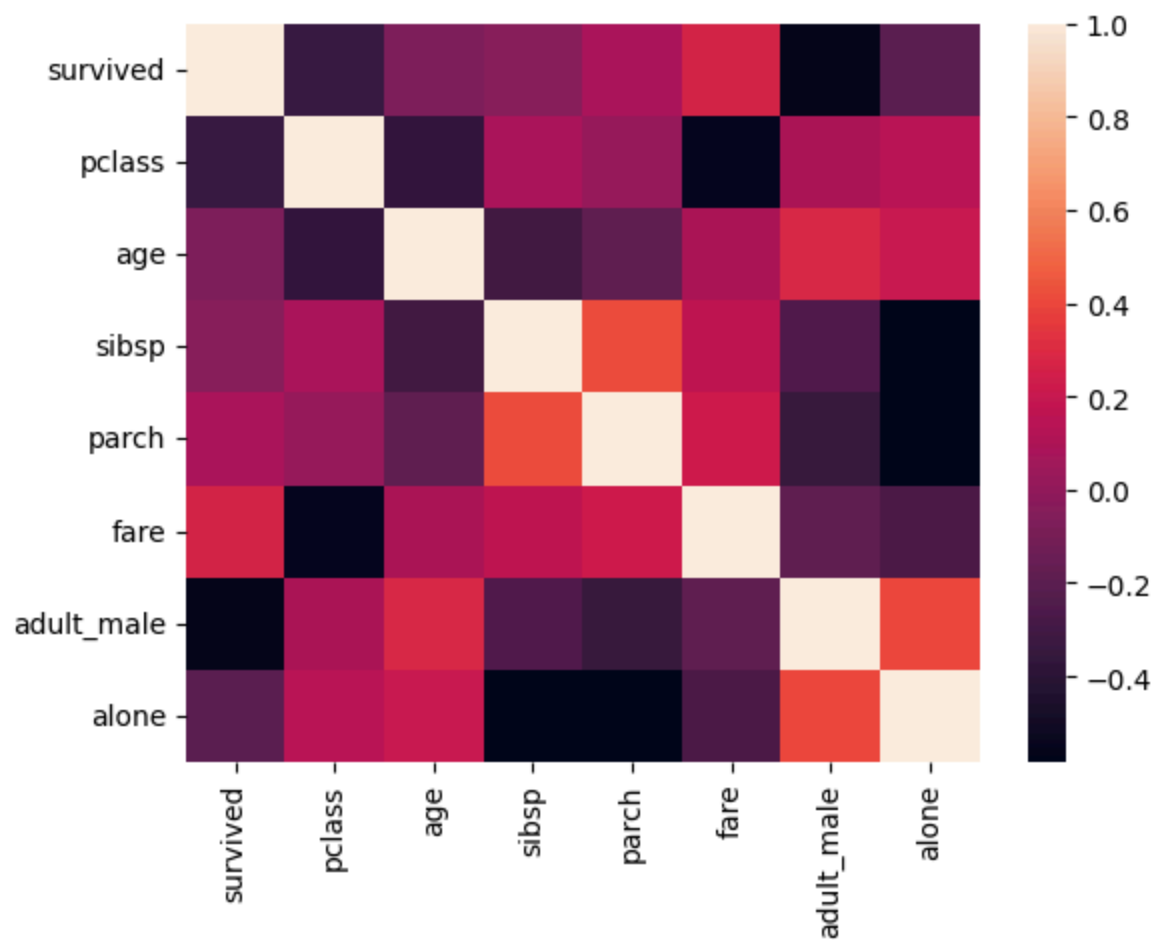
```
In [19]: titanic.corr(numeric_only= True)
```

```
Out[19]:
```

	survived	pclass	age	sibsp	parch	fare	adult
survived	1.000000	-0.338481	-0.077221	-0.035322	0.081629	0.257307	-0.5
pclass	-0.338481	1.000000	-0.369226	0.083081	0.018443	-0.549500	0.0
age	-0.077221	-0.369226	1.000000	-0.308247	-0.189119	0.096067	0.2
sibsp	-0.035322	0.083081	-0.308247	1.000000	0.414838	0.159651	-0.2
parch	0.081629	0.018443	-0.189119	0.414838	1.000000	0.216225	-0.3
fare	0.257307	-0.549500	0.096067	0.159651	0.216225	1.000000	-0.1
adult_male	-0.557080	0.094035	0.280328	-0.253586	-0.349943	-0.182024	1.0
alone	-0.203367	0.135207	0.198270	-0.584471	-0.583398	-0.271832	0.4

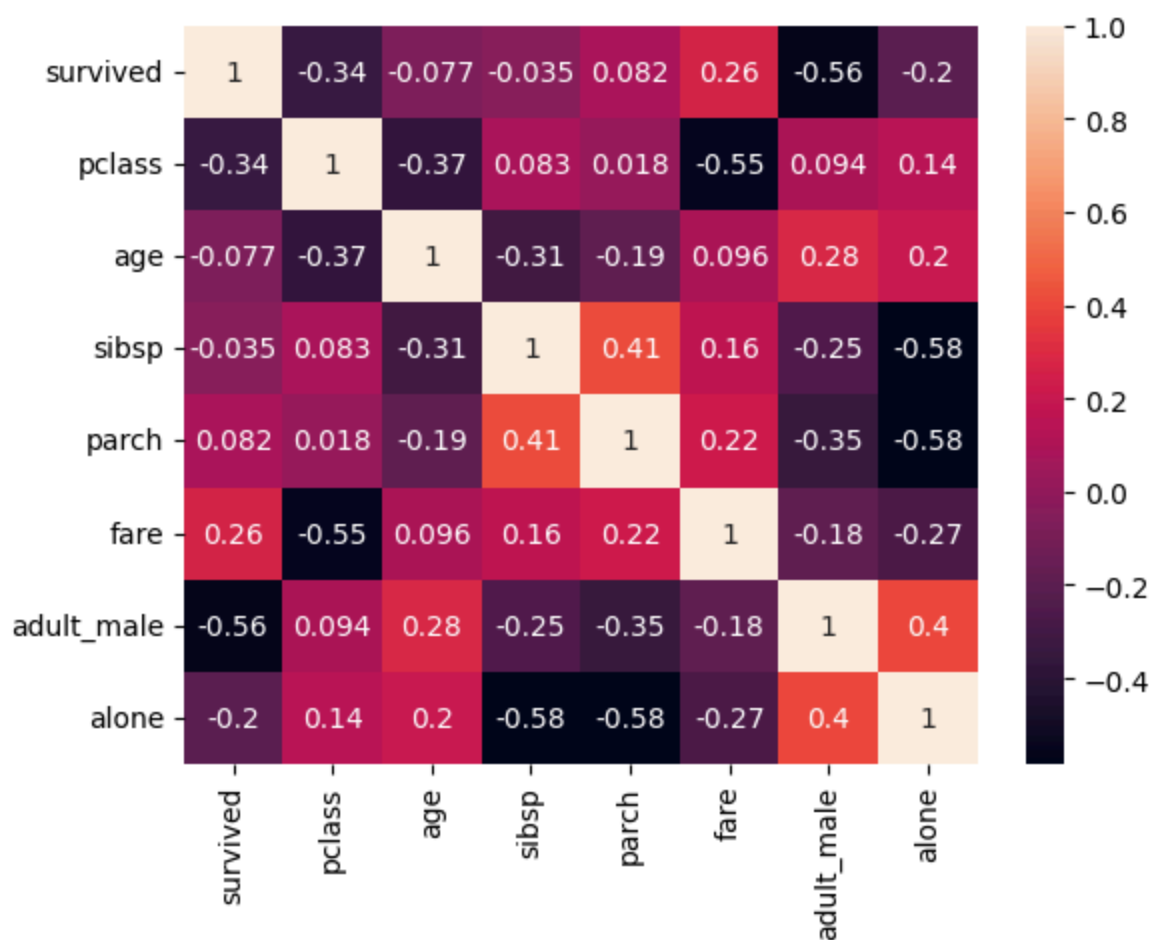
```
In [20]: sns.heatmap(titanic.corr(numeric_only= True))
```

```
Out[20]: <Axes: >
```



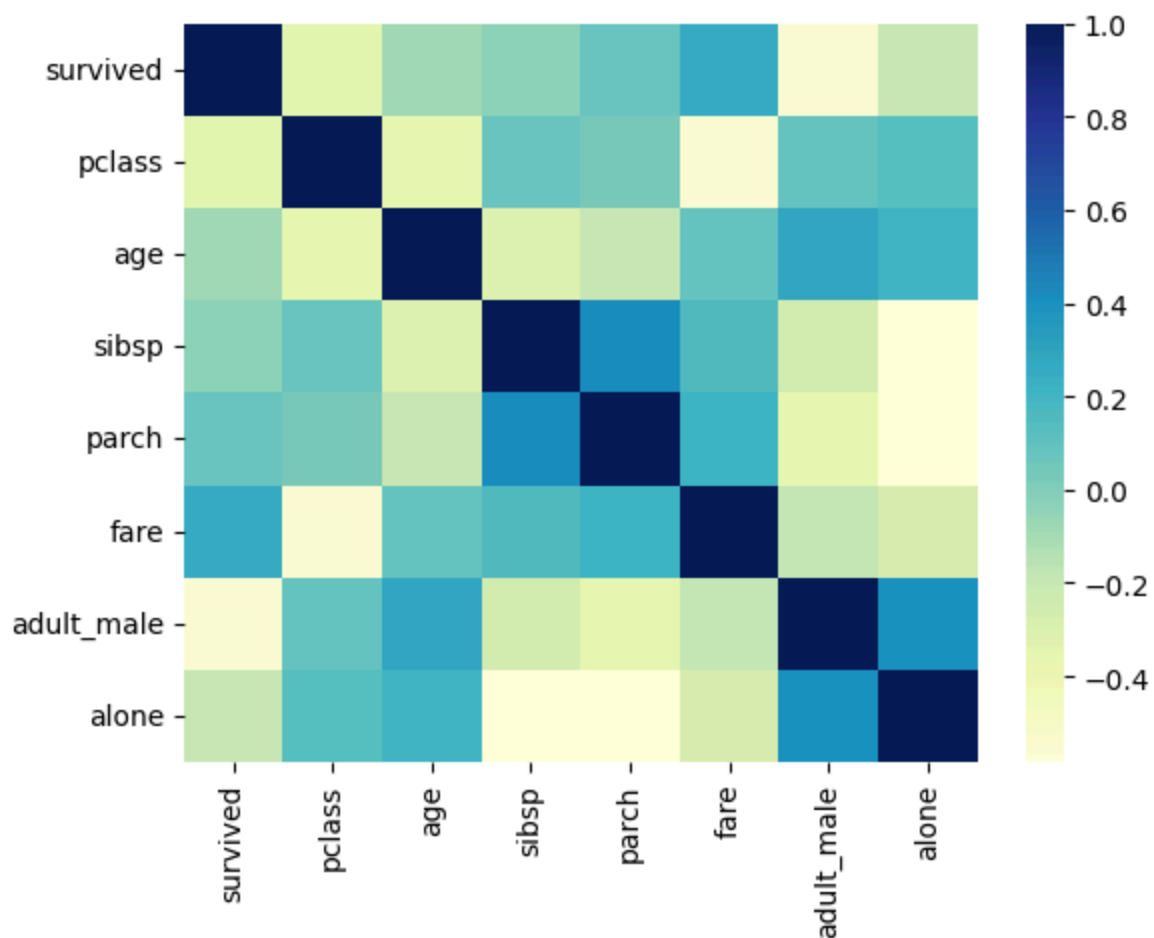
```
In [21]: sns.heatmap(titanic.corr(numeric_only=True), annot=True)
```

```
Out[21]: <Axes: >
```

```
In [22]: sns.heatmap(titanic.corr(numeric_only=True), cmap = 'YlGnBu')
```

```
Out[22]: <Axes: >
```



```
In [23]: plt.hist(x = titanic['fare'], bins = 20)
```

```
Out[23]: (array([562., 170., 67., 39., 15., 16., 2., 0., 9., 2., 6.,
0., 0., 0., 0., 0., 0., 0., 0., 3.]),
array([ 0., 25.61646, 51.23292, 76.84938, 102.46584, 128.0823 ,
153.69876, 179.31522, 204.93168, 230.54814, 256.1646 , 281.78106,
307.39752, 333.01398, 358.63044, 384.2469 , 409.86336, 435.47982,
461.09628, 486.71274, 512.3292 ]),
<BarContainer object of 20 artists>)
```

